

Module - 03

Network Layer-Part II

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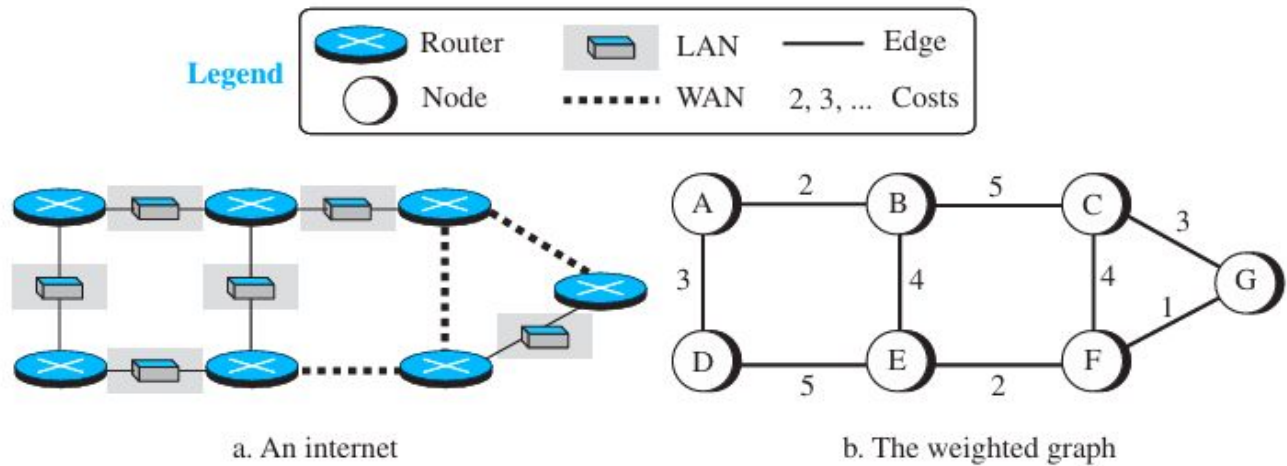
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3.1.Unicast Routing : Routing Algorithms

- In an internet, the goal of the network layer is to deliver a datagram from its source to its destination or destinations. If a datagram is destined for only one destination (one-to-one delivery), we have **unicast routing**. If the datagram is destined for several destinations (one-to-many delivery), we have **multicast routing**.

Figure 20.1 *An internet and its graphical representation*

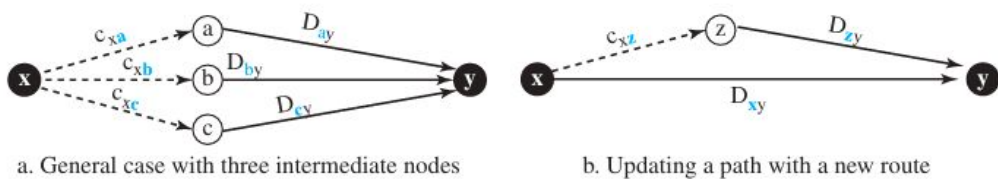


3.1.1. Distance-Vector Routing

Bellman-Ford Equation

- $D_{xy} = \min\{(c_{xa} + D_{ay}), (c_{xb} + D_{by}), (c_{xc} + D_{cy}), \dots\}$
- $D_{xy} = \min\{D_{xy}, (c_{xz} + D_{zy})\}$

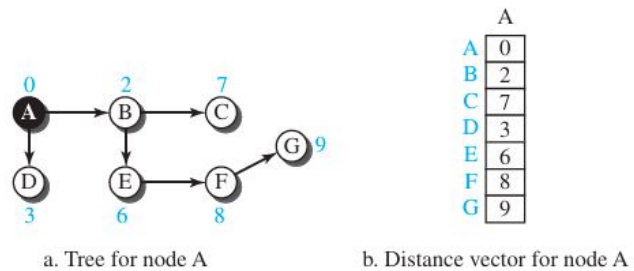
Figure 20.3 Graphical idea behind Bellman-Ford equation



Distance Vectors

- Figure 20.4 shows the tree for node A in the internet in Figure 20.1 and the corresponding distance vector.

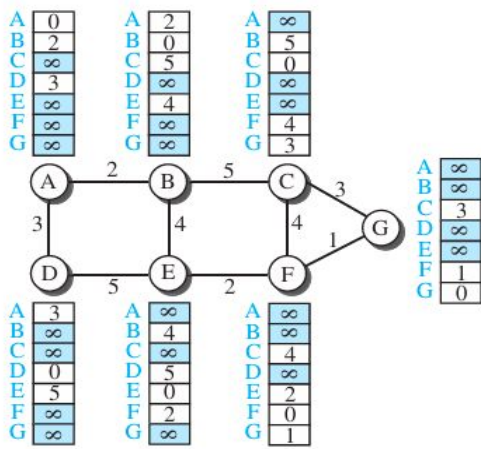
Figure 20.4 The distance vector corresponding to a tree



We know that a distance vector can represent least-cost paths in a least-cost tree, but the question is **how each node in an internet originally created the corresponding vector.**

•

Figure 20.5 *The first distance vector for an internet*



Each node in an internet, when it is booted, creates a very rudimentary(basic) distance vector with the minimum information the node can obtain from its neighborhood.

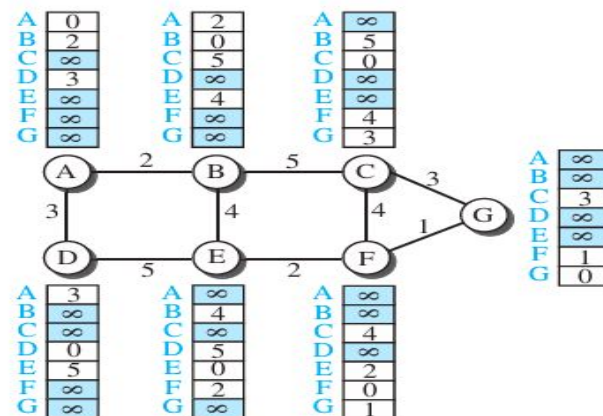
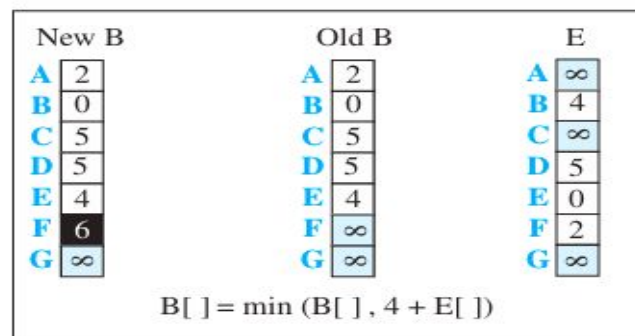
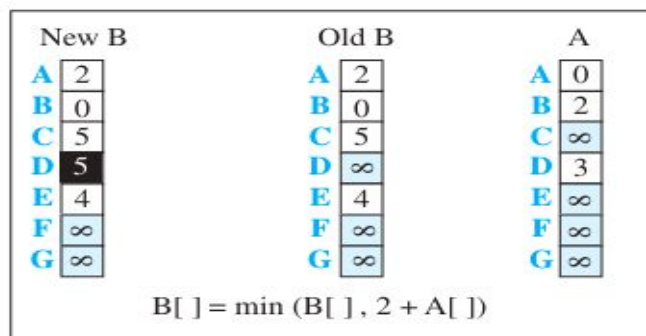


Figure 20.6 Updating distance vectors



Note:
X[]: the whole vector

Distance-Vector Routing Algorithm

- Now we can give a simplified pseudocode for the distance-vector routing algorithm. The algorithm is run by its node independently and asynchronously.
- **Distance vector Routing Algorithm**

```
1 Distance_Vector_Routing ( )
2 {
    3 // Initialize (create initial vectors for the node)
    4 D[myself] = 0
    5 for (y = 1 to N)
    6 {
        7 if (y is a neighbor)
        8 D[y] = c[myself][y]
        9 else
        10 D[y] = ∞
    11 }
    12 send vector {D[1], D[2], ..., D[N]} to all neighbors
    13 // Update (improve the vector with the vector received from a neighbor)
    14 repeat (forever)
    15 {
        16 wait (for a vector Dw from a neighbor w or any change in the link)
        17 for (y = 1 to N)
        18 {
            19 D[y] = min [D[y], (c[myself][w] + Dw[y])] // Bellman-Ford equation
        20 }
        21 if (any change in the vector)
        22 send vector {D[1], D[2], ..., D[N]} to all neighbors
    23 }
    24 } // End of Distance Vector
```

Steps	Action	Explanation
1	Initialize	Know your cost to all nodes (neighbors or ∞)
2	Send Initial Vector	Tell your neighbours your distances
3	Receives Update	Get other routers distance vectors
4	Apply Bellman ford	Update Your table using neighbours info
5	Send Updates again	If anything improved tell your neighbours
6	Repeat forever	Always keep the routing table fresh

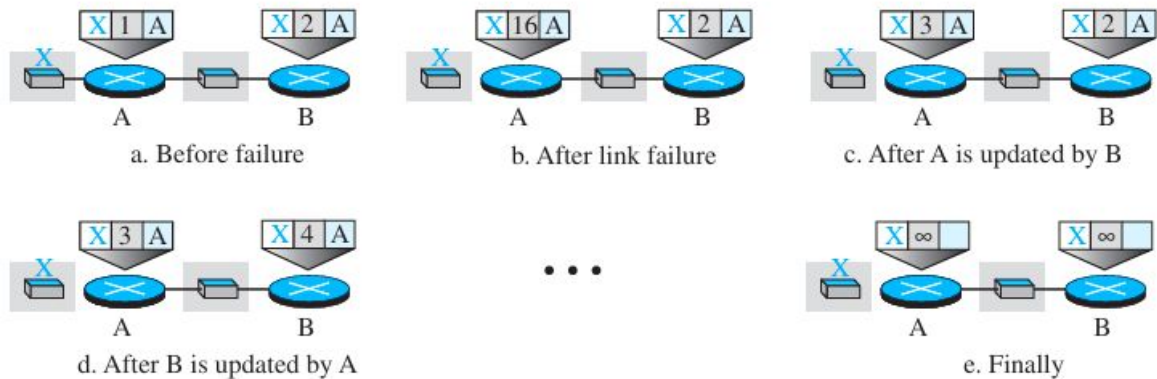
Count to Infinity

- A problem with distance-vector routing is that any decrease in cost (good news) propagates quickly, but any increase in cost (bad news) will propagate slowly. For a routing protocol to work properly, if a link is broken (cost becomes infinity), every other router should be aware of it immediately, but in distance-vector routing, this takes some time.
- The problem is referred to as count to infinity. It sometimes takes several updates before the cost for a broken link is recorded as infinity by all routers.

Two-Node Loop

- At the beginning, both nodes A and B know how to reach node X. But suddenly, the link between A and X fails. Node A changes its table. If A can send its table to B immediately, everything is fine.
- However, the system becomes unstable if B sends its forwarding table to A before receiving A's forwarding table. Node A receives the update and, assuming that B has found a way to reach X, immediately updates its forwarding table. Now A sends its new update to B. Now B thinks that something has been changed around A and updates its forwarding table.
- The cost of reaching X increases gradually until it reaches infinity. At this moment, both A and B know that X cannot be reached.
- If A receives a packet destined for X, the packet goes to B and then comes back to A. Similarly, if B receives a packet destined for X, it goes to A and comes back to B. Packets bounce between A and B, creating a **two-node loop problem**.

Figure 20.7 Two-node instability



Split Horizon

- In this strategy, instead of flooding the table through each interface, each node sends only part of its table through each interface. If, according to its table, node B thinks that the optimum route to reach X is via A, it does not need to advertise this piece of information to A; the information has come from A (A already knows). Taking information from node A, modifying it, and sending it back to node A is what creates the confusion.

Poison Reverse

- In the poison reverse strategy B can still advertise the value for X, but if the source of information is A, it can replace the distance with infinity as a warning: “Do not use this value; what I know about this route comes from you.”

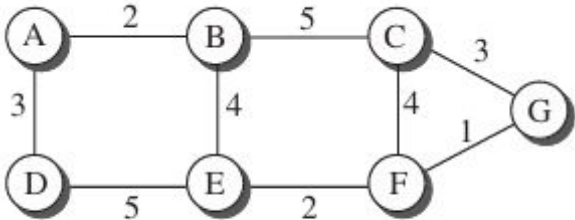
Three-Node Instability

- The two-node instability can be avoided using split horizon combined with poison reverse. However, if the instability is between three nodes, stability cannot be guaranteed

3.1.2. Link State Routing

- A routing algorithm that directly follows least-cost trees and forwarding tables is link-state (LS) routing.
- This method uses the term link-state to define the characteristic of a link (an edge) that represents a network in the internet.
- In this algorithm the cost associated with an edge defines the state of the link.
- Links with lower costs are preferred to links with higher costs; if the cost of a link is infinity, it means that the link does not exist or has been broken.

Figure 20.8 *Example of a link-state database*



a. The weighted graph

	A	B	C	D	E	F	G
A	0	2	∞	3	∞	∞	∞
B	2	0	5	∞	4	∞	∞
C	∞	5	0	∞	∞	4	3
D	3	∞	∞	0	5	∞	∞
E	∞	4	∞	5	0	2	∞
F	∞	∞	4	∞	2	0	1
G	∞	∞	3	∞	∞	1	0

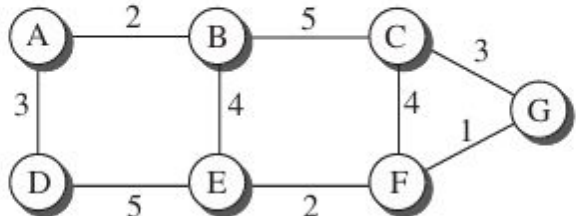
b. Link state database

3.1.2. Link State Routing

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Link-State Database (LSDB)

Figure 20.8 *Example of a link-state database*



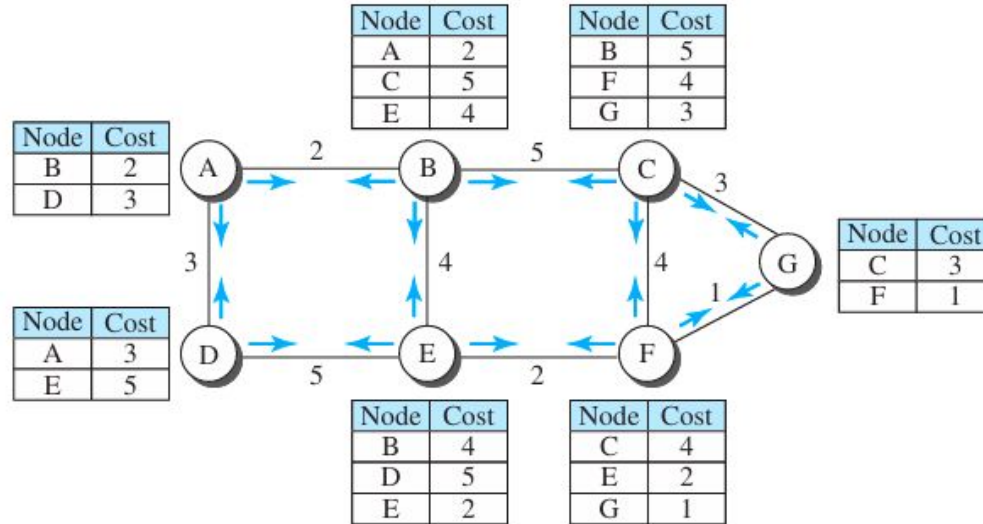
a. The weighted graph

	A	B	C	D	E	F	G
A	0	2	∞	3	∞	∞	∞
B	2	0	5	∞	4	∞	∞
C	∞	5	0	∞	∞	4	3
D	3	∞	∞	0	5	∞	∞
E	∞	4	∞	5	0	2	∞
F	∞	∞	4	∞	2	0	1
G	∞	∞	3	∞	∞	1	0

b. Link state database

- The LSP is sent out of each interface, as shown in Figure 20.9 for our internet.

Figure 20.9 *LSPs created and sent out by each node to build LSDB*



Formation of Least-Cost Trees

To create a least-cost tree for itself, using the shared LSDB, each node needs to run the famous Dijkstra Algorithm.

- **Dijkstra's Algorithm**

Dijkstra's Algorithm ()

```
{ // Initialization
    Tree = {root} // Tree is made only of the root
    for (y = 1 to N)
    {
        if (y is the root) D[y] = 0 else if (y is a neighbor) D[y] = c[root][y] else D[y] = ∞
    }
    // Calculation repeat
    { find a node w, with D[w] minimum among all nodes not in the Tree Tree = Tree ∪ {w}
        // Update distances for all neighbors of w
        for (every node x, which is a neighbor of w and not in the Tree)
        {
            D[x] = min {D[x], (D[w] + c[w][x])} }
        } until (all nodes included in the Tree)
    }
    // End of Dijkstra
```

Path Vector Routing

3.2. Unicast Routing Protocols

- A protocol needs to define its domain of operation, the messages exchanged, communication between routers, and interaction with protocols in other domains

Internet Structure

Hierarchical Routing

Autonomous System

a group of interconnected IP networks and routers under a single administrative control that shares a common routing policy

We have stub ASs, multihomed ASs, and transient ASs.

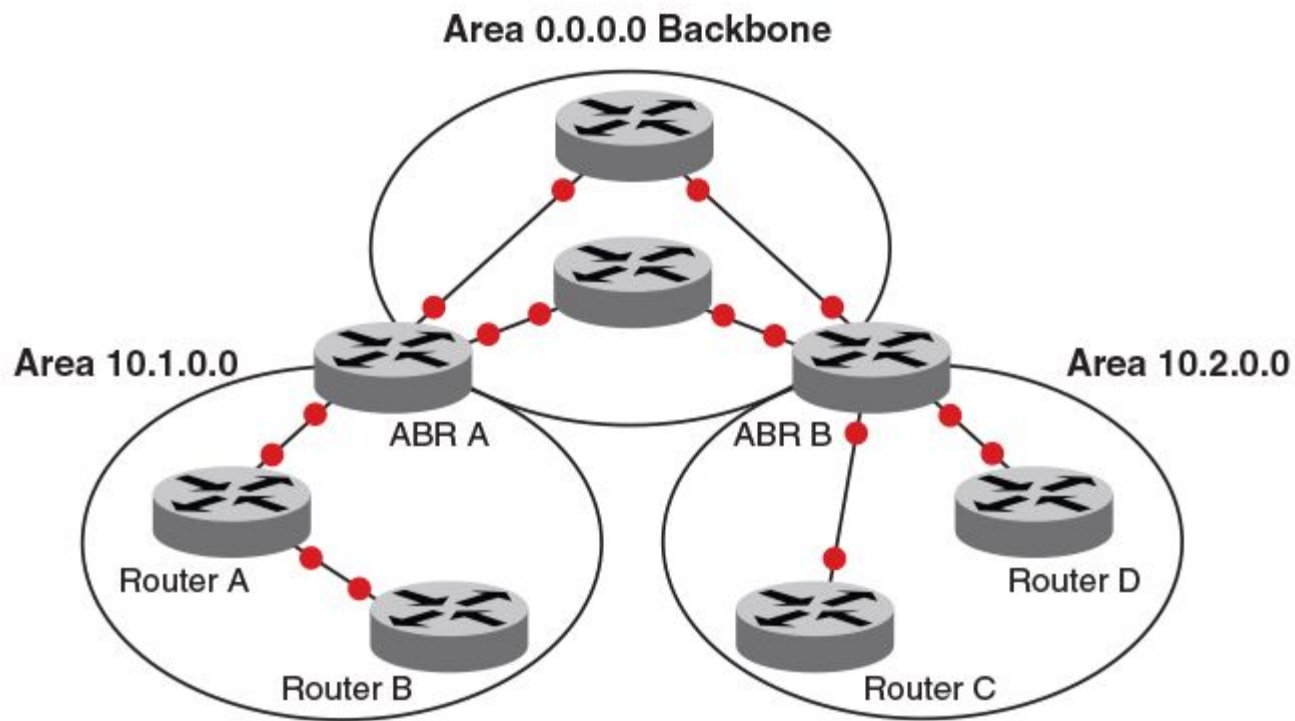


Figure 20.15 *Hop counts in RIP*

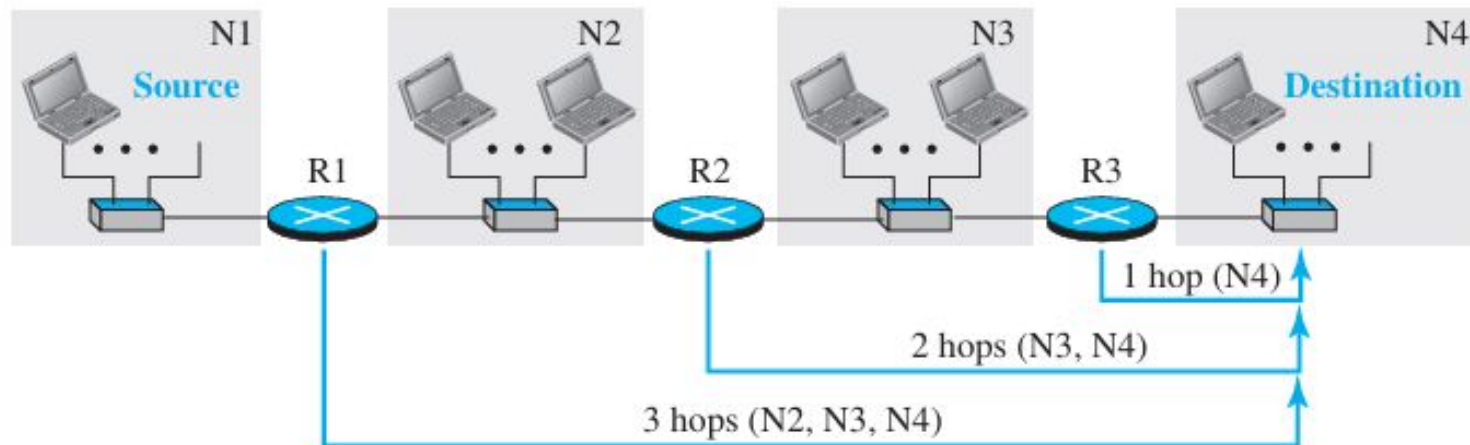
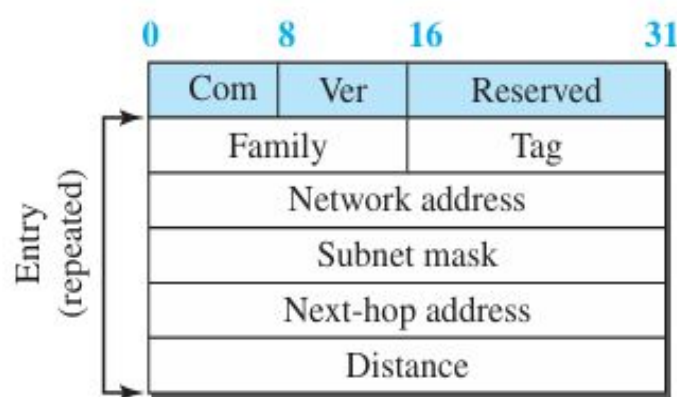


Figure 20.16 *Forwarding tables*

Forwarding table for R1			Forwarding table for R2			Forwarding table for R3		
Destination network	Next router	Cost in hops	Destination network	Next router	Cost in hops	Destination network	Next router	Cost in hops
N1	—	1	N1	R1	2	N1	R2	3
N2	—	1	N2	—	1	N2	R2	2
N3	R2	2	N3	—	1	N3	—	1
N4	R2	3	N4	R3	2	N4	—	1

Figure 20.17 *RIP message format*



Fields

Com: Command, request (1), response (2)

Ver: Version, current version is 2

Family: Family of protocol, for TCP/IP value is 2

Tag: Information about autonomous system

Network address: Destination address

Subnet mask: Prefix length

Next-hop address: Address length

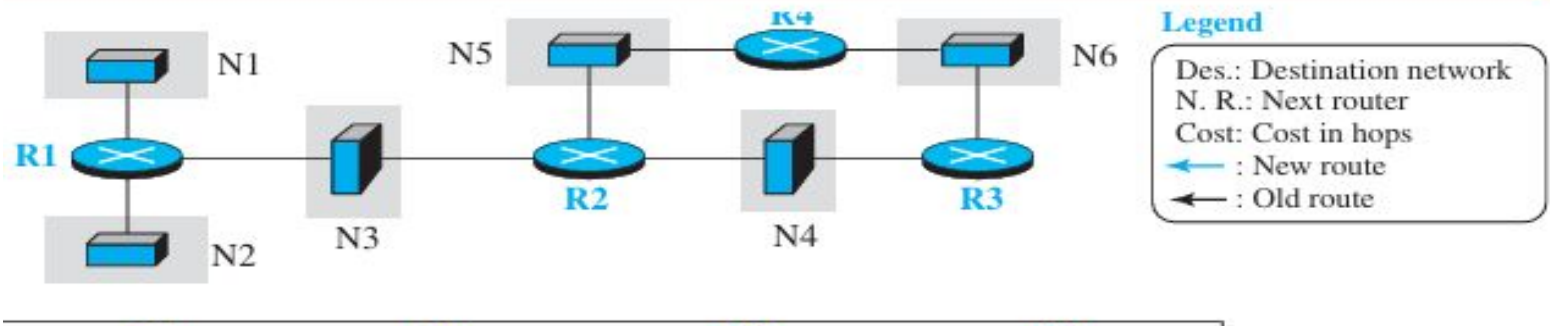
Distance: Number of hops to the destination

RIP Algorithm

- Instead of sending only distance vectors, a router needs to send the whole contents of its forwarding table in a response message.
- The receiver adds one hop to each cost and changes the next router field to the address of the sending router. We call each route in the modified forwarding table the received route and each route in the old forwarding table the old route. The received router selects the old routes as the new ones except in the following three cases:
 - 1. If the received route does not exist in the old forwarding table, it should be added to the route.
 - 2. If the cost of the received route is lower than the cost of the old one, the received route should be selected as the new one.
 - 3. If the cost of the received route is higher than the cost of the old one, but the value of the next router is the same in both routes, the received route should be selected as the new one. This is the case where the route was actually advertised by the same router in the past, but now the situation has been changed. For example, suppose a neighbor has previously advertised a route to a destination with cost 3, but now there is no path between this neighbor and that destination. The neighbor advertises this destination with cost value infinity (16 in RIP). The receiving router must not ignore this value even though its old route has a lower cost to the same destination.
- The new forwarding table needs to be sorted according to the destination route (mostly using the longest prefix first).

Q1) Construct the forwarding tables for all the routers after they have been stabilized

Figure 20.18 Example of an autonomous system using RIP



Forwarding tables after all routers booted

R1			R2			R3			R4		
Des.	N. R.	Cost	Des.	N. R.	Cost	Des.	N. R.	Cost	Des.	N. R.	Cost
N1	_____	1	N3	_____	1	N4	_____	1	N5	_____	1
N2	_____	1	N4	_____	1	N6	_____	1	N6	_____	1
N3	_____	1	N5	_____	1						

Changes in the forwarding tables of R1, R3, and R4 after they receive a copy of R2's table

New R1			Old R1			R2 Seen by R1		
Des.	N. R.	Cost	Des.	N. R.	Cost	Des.	N. R.	Cost
N1	_____	1	N1	_____	1	N3	R2	2
N2	_____	1	N2	_____	1	N4	R2	2
N3	_____	1	N3	_____	1	N5	R2	2
N4	R2	2						
N5	R2	2						

New R3			Old R3			R2 Seen by R3		
Des.	N. R.	Cost	Des.	N. R.	Cost	Des.	N. R.	Cost
N3	R2	2	N4	_____	1	N3	R2	2
N4	_____	1	N6	_____	1	N4	R2	2
N5	R2	2				N5	R2	2
N6	_____	1						

New R4			Old R4			R2 Seen by R4		
Des.	N. R.	Cost	Des.	N. R.	Cost	Des.	N. R.	Cost
N3	R2	2	N5	_____	1	N3	R2	2
N4	R2	2	N6	_____	1	N4	R2	2
N5	_____	1				N5	R2	2
N6	_____	1						

Forwarding tables for all routers after they have been stabilized

Final R1			Final R2			Final R3			Final R4		
Des.	N. R.	Cost	Des.	N. R.	Cost	Des.	N. R.	Cost	Des.	N. R.	Cost
N1	_____	1	N1	R1	2	N1	R2	3	N1	R2	3
N2	_____	1	N2	R1	2	N2	R2	3	N2	R2	3
N3	_____	1	N3	_____	1	N3	R2	2	N3	R2	2
N4	R2	2	N4	_____	1	N4	_____	1	N4	R2	2
N5	R2	2	N5	_____	1	N5	R2	2	N5	_____	1
N6	R2	3	N6	R3	2	N6	_____	1	N6	_____	1

Step-by-Step Explanation

- **1. At Boot (first row of tables)**
 - A. **R1** knows only N1, N2, N3 (directly connected).
 - B. **R2** knows only N3, N4, N5.
 - C. **R3** knows only N4, N6.
 - D. **R4** knows only N5, N6.
- They don't yet know how to reach networks not directly attached.
- **2. Exchange of Updates**
- RIP periodically sends its whole routing table to neighbors.
- Example
 - A. **R2** sends its table to R1.
 - B. R1 sees: “Oh, R2 can reach N4 and N5 with cost 1.
So for me, via R2 it would cost 2.”
 - C. R1 adds those entries in its table:
N4 → via R2 cost 2, N5 → via R2 cost 2.
- Same for R3 and R4:
 - A. R3 receives from R2 and learns about N3, N5.
 - B. R4 receives from R2 and learns about N3, N4
- This is exactly what the “New R1 / Old R1 / R2 Seen by R1” boxes show — the new routes learned.

- **3. After Several Rounds**
- All routers keep exchanging updates until no table changes anymore — this is called **convergence**.
- In the **final tables** at the bottom:
- Each router now knows how to reach **all six networks**.
- They all have the **shortest path (lowest hop count)** recorded.
- Example from Final R1:
 - A. N1, N2, N3 directly connected (cost 1).
 - B. N4 via R2 (cost 2).
 - C. N5 via R2 (cost 2).
 - D. N6 via R2 (cost 3).

Timers and Performance in RIP

3.2.2. Open Shortest Path First

- Open Shortest Path First (OSPF) is also an intradomain routing protocol like RIP, but it is based on the link-state routing protocol.
- OSPF is an open protocol, which means that the specification is a public document.

Metric(mathematics)

Figure 20.19 *Metric in OSPF*

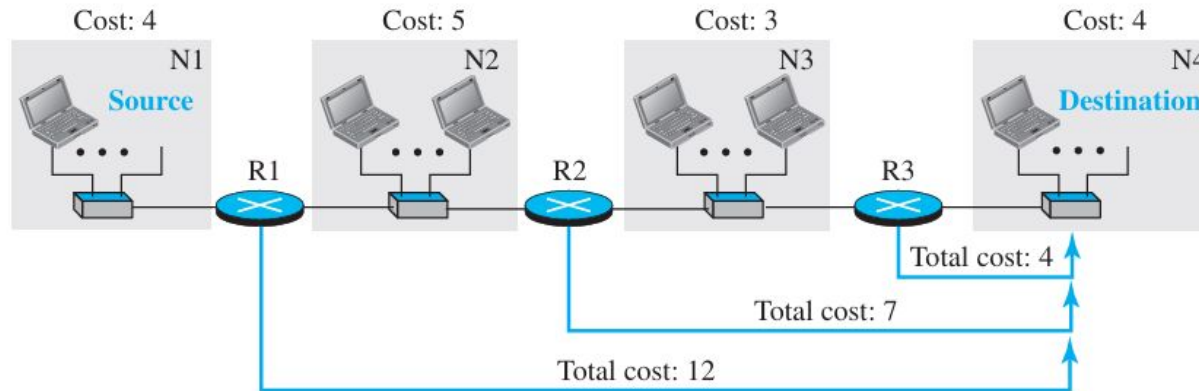


Figure 20.19 *Metric in OSPF*

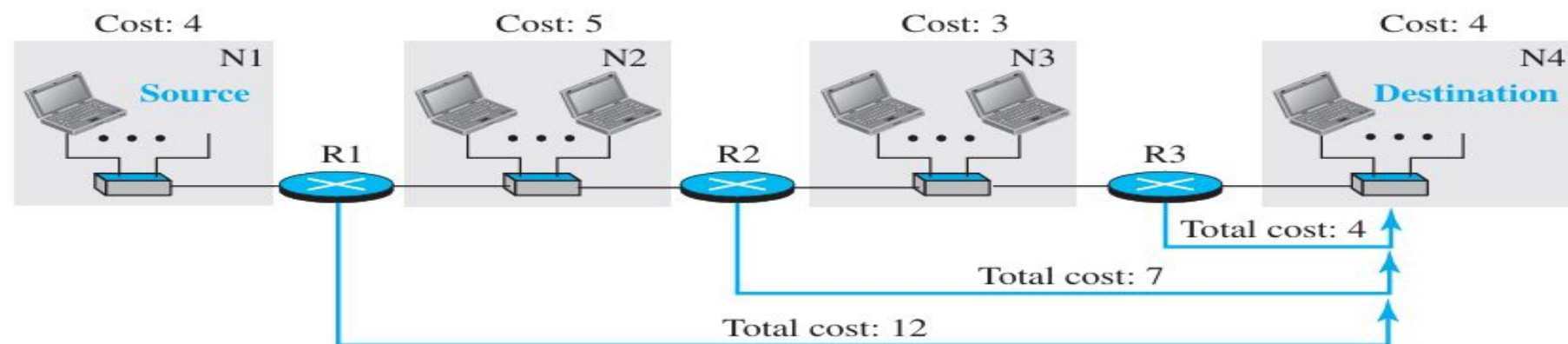
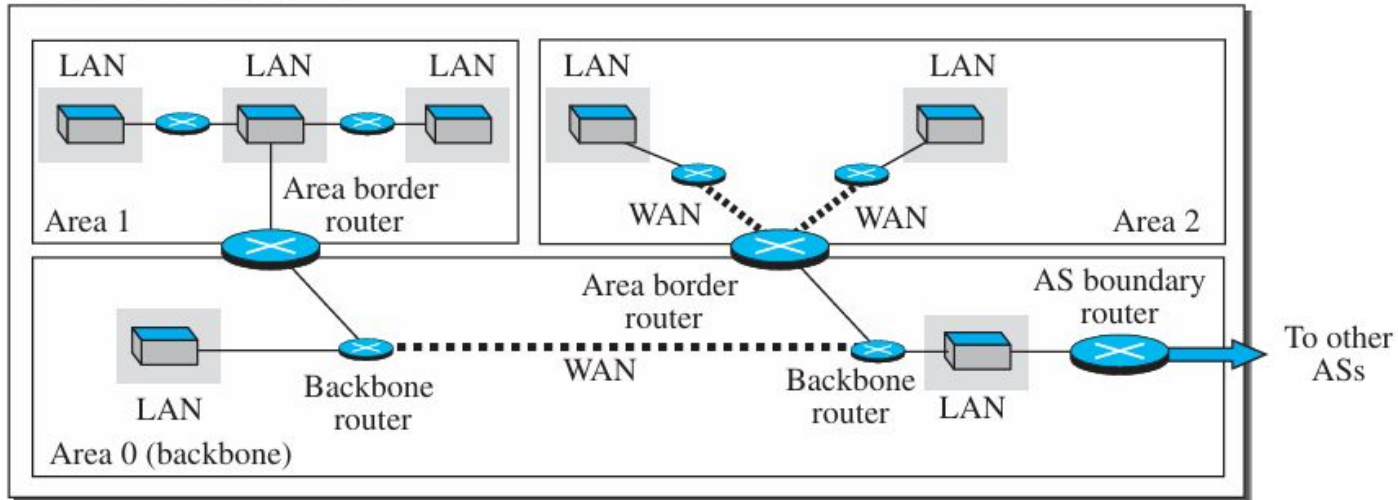


Figure 20.20 *Forwarding tables in OSPF*

Forwarding table for R1			Forwarding table for R2			Forwarding table for R3		
Destination network	Next router	Cost	Destination network	Next router	Cost	Destination network	Next router	Cost
N1	—	4	N1	R1	9	N1	R2	12
N2	—	5	N2	—	5	N2	R2	8
N3	R2	8	N3	—	3	N3	—	3
N4	R2	12	N4	R3	7	N4	—	4

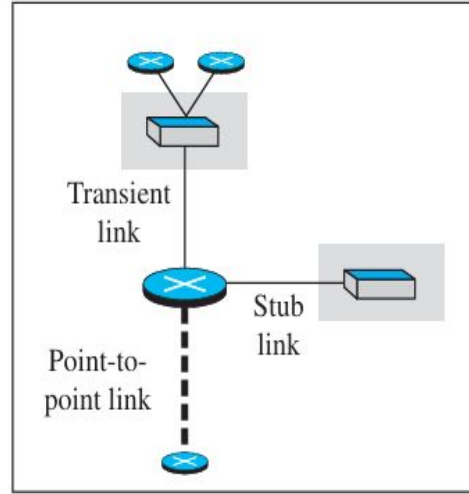
Figure 20.21 *Areas in an autonomous system*

Autonomous System (AS)

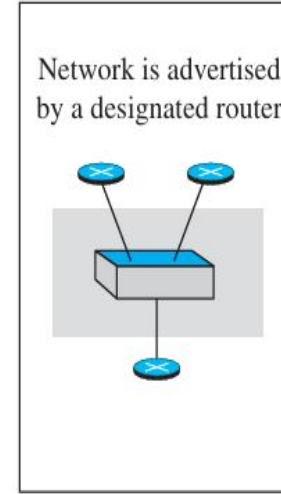


Link-State Advertisement

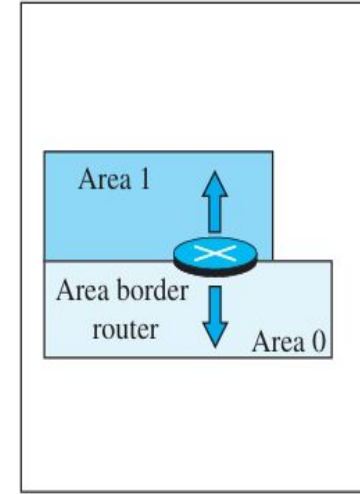
- We have 5 types of link state advertisements:
 - Router link
 - Network link
 - Summary link to network
 - Summary link to AS border router
 - External link.



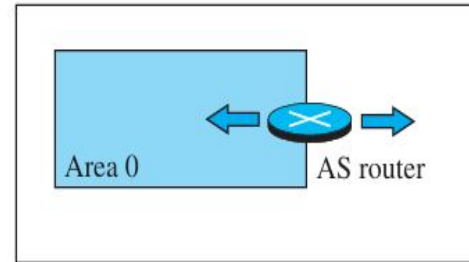
a. Router link



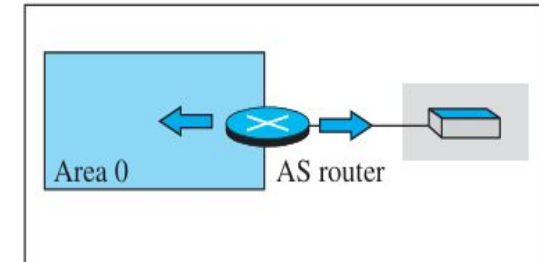
b. Network link



c. Summary link to network

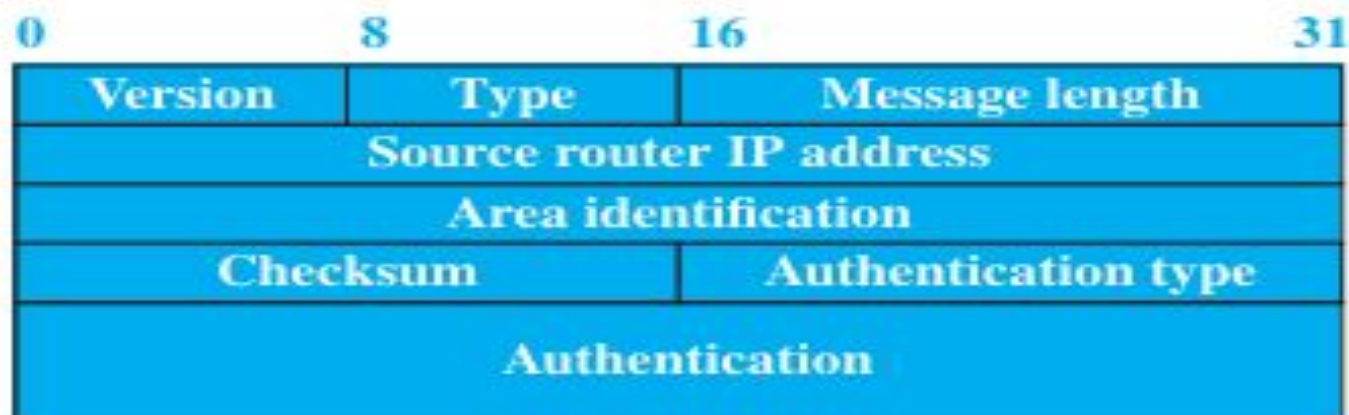


d. Summary link to AS



e. External link

Figure 20.23 *OSPF message formats*



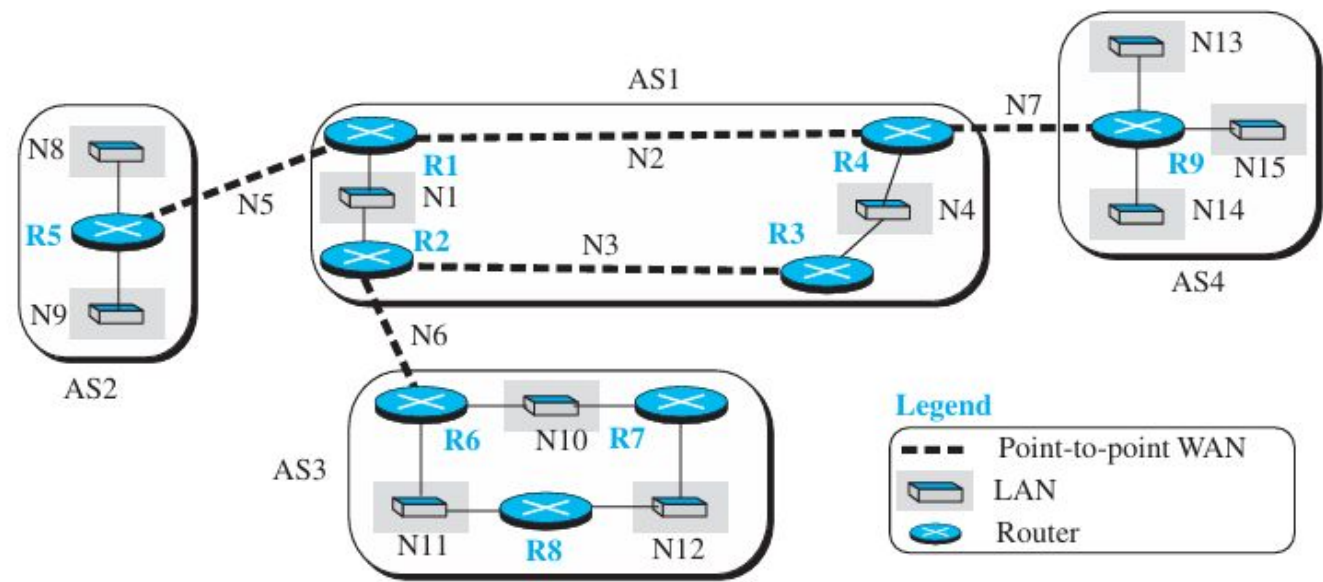
OSPF common header

- **OSPF Algorithm** OSPF implements the link-state routing algorithm we discussed in the previous section. However, some changes and augmentations need to be added to the algorithm:
 - After each router has created the shortest-path tree, the algorithm needs to use it to create the corresponding routing algorithm.
 - The algorithm needs to be augmented to handle sending and receiving all five types of messages.

3.2.3.Border Gateway Protocol Version 4 (BGP4)

- Figure 20.24 shows an example of an internet with four autonomous systems. AS2, AS3, and AS4 are stub autonomous systems; AS1 is a transient one. In our example, data exchange between AS2, AS3, and AS4 should pass through AS1.

Figure 20.24 *A sample internet with four ASs*

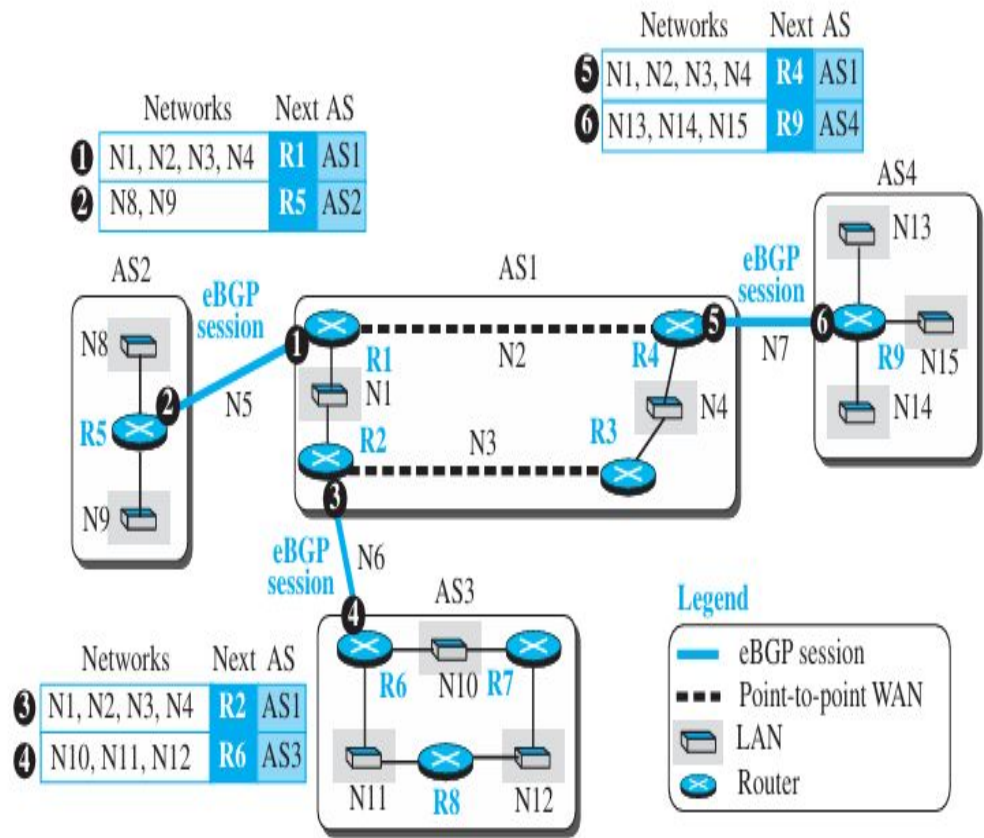


3.2.3.Border Gateway Protocol Version 4 (BGP4)

The figure also shows the simplified update messages sent by routers involved in the eBGP sessions. The circled number defines the sending router in each case.

- For example, message number 1 is sent by router R1 and tells router R5 that N1, N2, N3, and N4 can be reached through router R1 (R1 gets this information from the corresponding intradomain forwarding table).
- Router R5 can now add these pieces of information at the end of its forwarding table.
- When R5 receives any packet destined for these four networks, it can use its forwarding table and find that the next router is R1.

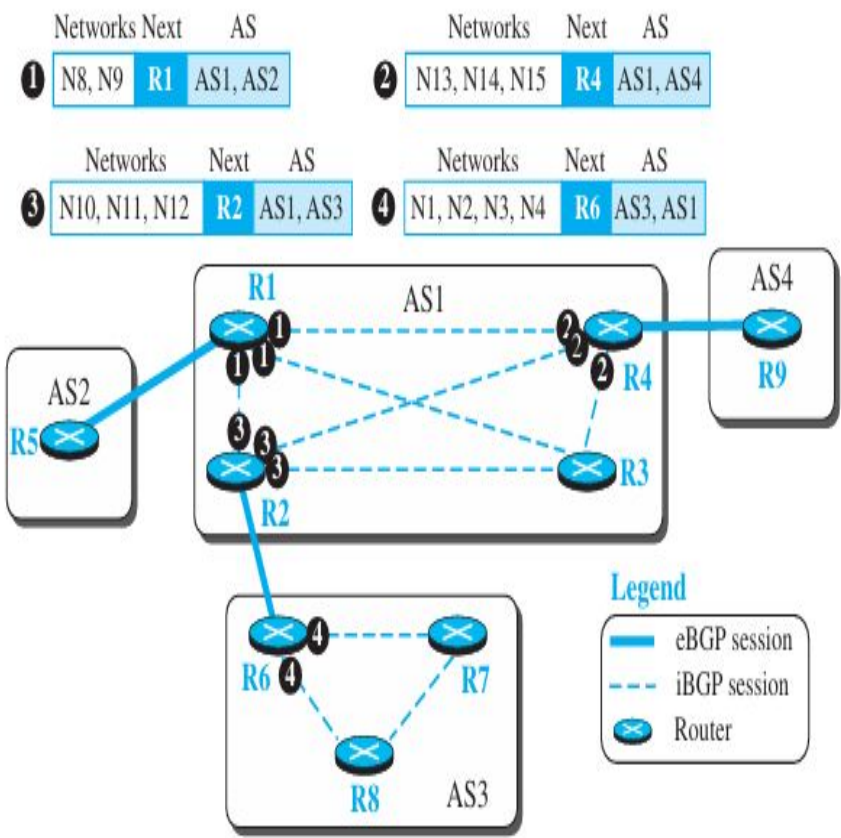
Figure 20.25 eBGP operation



3.2.3.Border Gateway Protocol Version 4 (BGP4)

- Note that in this stage only four messages are exchanged. The first message (numbered 1) is sent by R1 announcing that networks N8 and N9 are reachable through the path AS1-AS2, but the next router is R1. This message is sent, through separate sessions, to R2, R3, and R4.
- Routers R2, R4, and R6 do the same thing but send different messages to different destinations.
- The interesting point is that, at this stage, R3, R7, and R8 create sessions with their peers, but they actually have no message to send.

Figure 20.26 Combination of eBGP and iBGP sessions in our internet



- we show the path tables in Figure 20.27 for the routers in Figure 20.24.

Figure 20.27 *Finalized BGP path tables*

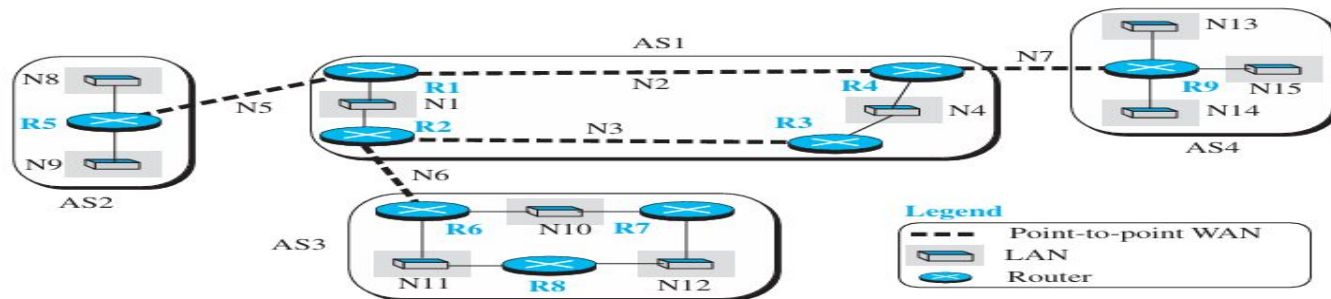
Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N8, N9	R5	AS1, AS2	N8, N9	R1	AS1, AS2	N8, N9	R2	AS1, AS2
N10, N11, N12	R2	AS1, AS3	N10, N11, N12	R6	AS1, AS3	N10, N11, N12	R2	AS1, AS3
N13, N14, N15	R4	AS1, AS4	N13, N14, N15	R1	AS1, AS4	N13, N14, N15	R4	AS1, AS4

Path table for R1			Path table for R2			Path table for R3		
Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N8, N9	R1	AS1, AS2	N1, N2, N3, N4	R1	AS2, AS1	N1, N2, N3, N4	R2	AS3, AS1
N10, N11, N12	R1	AS1, AS3	N10, N11, N12	R1	AS2, AS1, AS3	N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R9	AS1, AS4	N13, N14, N15	R1	AS2, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4

Path table for R4			Path table for R5			Path table for R6		
Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R4	AS4, AS1
N8, N9	R6	AS3, AS1, AS2	N8, N9	R6	AS3, AS1, AS2	N8, N9	R4	AS4, AS1, AS2
N13, N14, N15	R6	AS3, AS1, AS4	N13, N14, N15	R6	AS3, AS1, AS4	N10, N11, N12	R4	AS4, AS1, AS3

Path table for R7			Path table for R8			Path table for R9		
Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R4	AS4, AS1
N8, N9	R6	AS3, AS1, AS2	N8, N9	R6	AS3, AS1, AS2	N8, N9	R4	AS4, AS1, AS2
N13, N14, N15	R6	AS3, AS1, AS4	N13, N14, N15	R6	AS3, AS1, AS4	N10, N11, N12	R4	AS4, AS1, AS3

- For example, router R1 now knows that any packet destined for networks N8 or N9 should go through AS1 and AS2 and the next router to deliver the packet to is router R5. Similarly, router R4 knows that any packet destined for networks N10, N11, or N12 should go through AS1 and AS3 and the next router to deliver this packet to is router R1, and so on.

Figure 20.24 *A sample internet with four ASs***Figure 20.27** *Finalized BGP path tables*

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N8, N9	R5	AS1, AS2	N8, N9	R1	AS1, AS2	N8, N9	R2	AS1, AS2
N10, N11, N12	R2	AS1, AS3	N10, N11, N12	R6	AS1, AS3	N10, N11, N12	R2	AS1, AS3
N13, N14, N15	R4	AS1, AS4	N13, N14, N15	R1	AS1, AS4	N13, N14, N15	R4	AS1, AS4

Path table for R1

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N8, N9	R1	AS1, AS2	N1, N2, N3, N4	R1	AS2, AS1	N1, N2, N3, N4	R2	AS3, AS1
N10, N11, N12	R1	AS1, AS3	N10, N11, N12	R1	AS2, AS1, AS3	N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R9	AS1, AS4	N13, N14, N15	R1	AS2, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4

Path table for R2

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R4	AS4, AS1
N8, N9	R6	AS3, AS1, AS2	N8, N9	R6	AS3, AS1, AS2	N8, N9	R4	AS4, AS1, AS2
N13, N14, N15	R6	AS3, AS1, AS4	N13, N14, N15	R6	AS3, AS1, AS4	N10, N11, N12	R4	AS4, AS1, AS3

Path table for R3

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1
N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4

Path table for R4

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R6	AS3, AS1	N1, N2, N3, N4	R4	AS4, AS1
N8, N9	R6	AS3, AS1, AS2	N8, N9	R6	AS3, AS1, AS2	N8, N9	R4	AS4, AS1, AS2
N13, N14, N15	R6	AS3, AS1, AS4	N13, N14, N15	R6	AS3, AS1, AS4	N10, N11, N12	R4	AS4, AS1, AS3

Path table for R5

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1
N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4

Path table for R6

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1
N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4

Path table for R7

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1
N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4

Path table for R8

Networks	Next	Path	Networks	Next	Path	Networks	Next	Path
N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1	N1, N2, N3, N4	R2	AS3, AS1
N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2	N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4	N13, N14, N15	R2	AS3, AS1, AS4

Path table for R9

Injection of Information into Intradomain Routing

- The role of an interdomain routing protocol such as BGP is to help the routers inside the AS to augment their routing information.
- **In the case of a stub AS**, the only area border router adds a default entry at the end of its forwarding table and defines the next router to be the speaker router at the end of the eBGP connection.
- **In the case of a transient AS**, the situation is more complicated. R1 in AS1 needs to inject the whole contents of the path table for R1 in Figure 20.27 into its intra domain forwarding table. The situation is the same for R2, R3, and R4.

Figure 20.24 A sample internet with four ASs

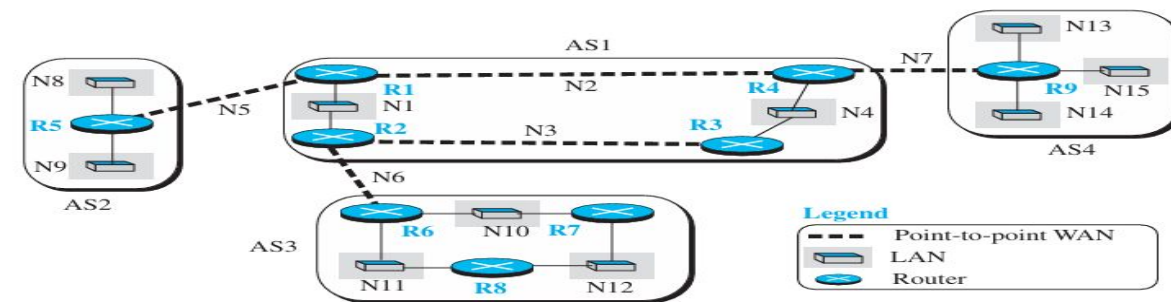


Figure 20.28 Forwarding tables after injection from BGP

Des.	Next	Cost
N1	—	1
N4	R4	2
N8	R5	1
N9	R5	1
N10	R2	2
N11	R2	2
N12	R2	2
N13	R4	2
N14	R4	2
N15	R4	2

Table for R1

Des.	Next	Cost
N1	—	1
N4	R3	2
N8	R1	2
N9	R1	2
N10	R6	1
N11	R6	1
N12	R6	1
N13	R3	3
N14	R3	3
N15	R3	3

Table for R2

Des.	Next	Cost
N1	R2	2
N4	—	1
N8	R2	3
N9	R2	3
N10	R2	2
N11	R2	2
N12	R2	2
N13	R4	2
N14	R4	2
N15	R4	2

Table for R3

Des.	Next	Cost
N1	R1	2
N4	—	1
N8	R1	2
N9	R1	2
N10	R3	3
N11	R3	3
N12	R3	3
N13	R9	1
N14	R9	1
N15	R9	1

Table for R4

Des.	Next	Cost
N8	—	1
N9	—	1
0	R1	1

Table for R5

Des.	Next	Cost
N10	—	1
N11	—	1
N12	R7	2
0	R2	1

Table for R6

Des.	Next	Cost
N10	—	1
N11	R6	2
N12	—	1
0	R6	2

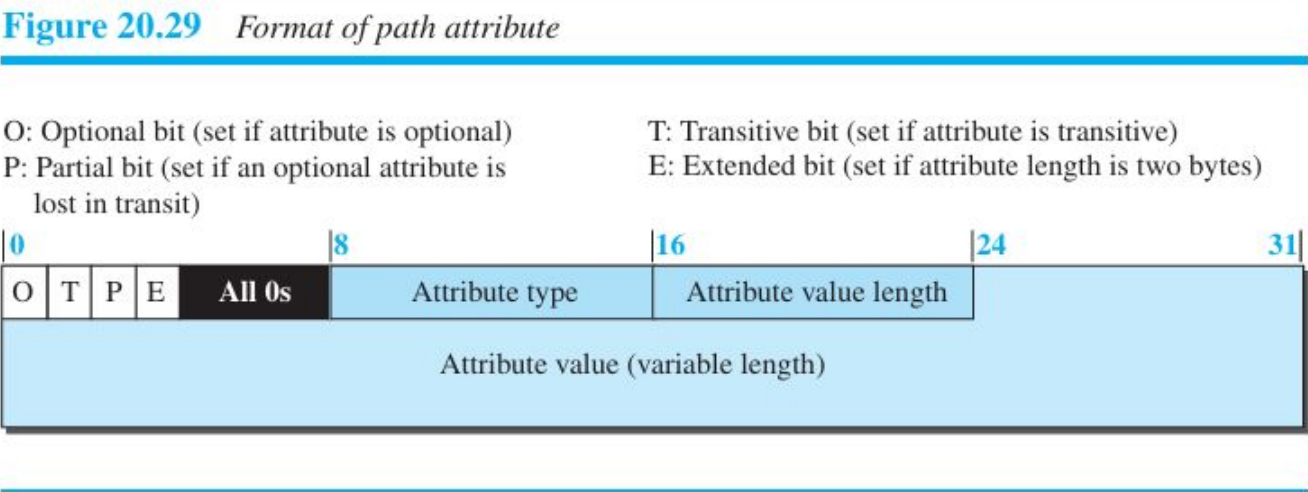
Table for R7

Des.	Next	Cost
N10	R6	2
N11	—	1
N12	—	1
0	R6	2

Table for R8

Des.	Next	Cost
N13	—	1
N14	—	1
N15	—	1
0	R4	1

Table for R9

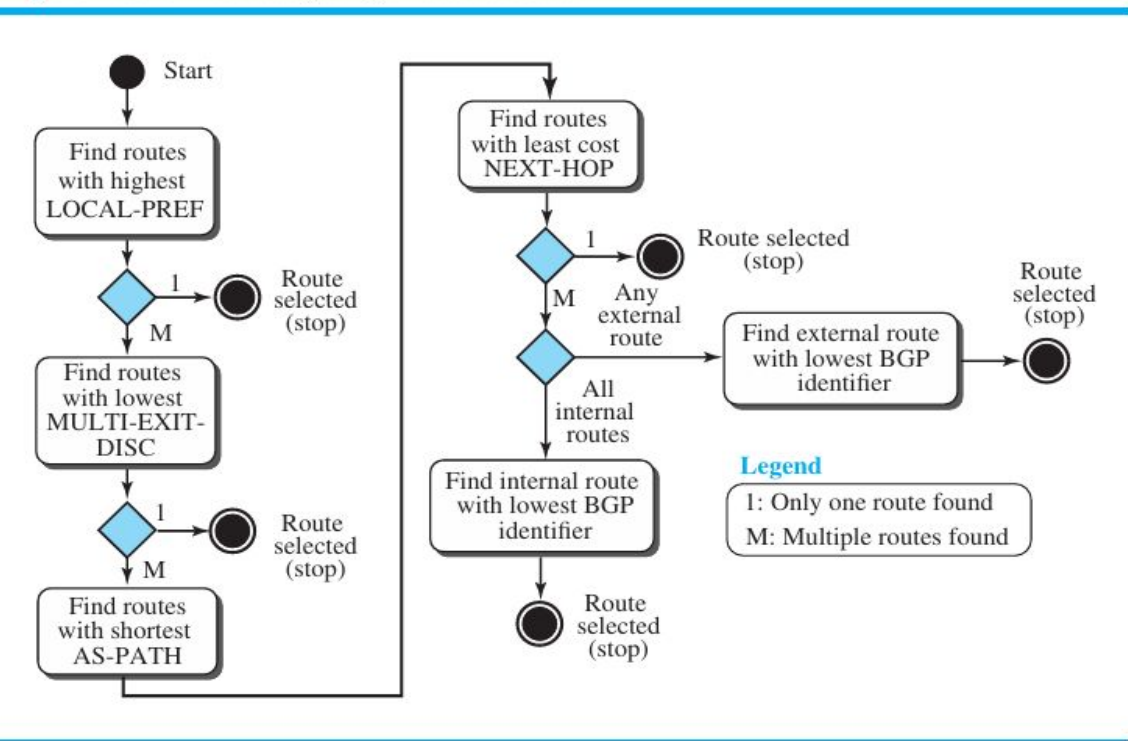


- The following gives a brief description of each attribute.
 - **ORIGIN (type 1).**
 - **AS-PATH (type 2).**
 - **NEXT-HOP (type 3).**
 - **MULT-EXIT-DISC (type 4).**
 - **LOCAL-PREF (type 5).**
 - **ATOMIC-AGGREGATE (type 6).**
 - **AGGREGATOR (type 7).**

Route Selection

- The router extracts the routes which meet the criteria in each step. If only one route is extracted, it is selected and the process stops; otherwise, the process continues with the next step. Note that the first choice is related to the LOCAL-PREF attribute, which reflects the policy imposed by the administration on the route

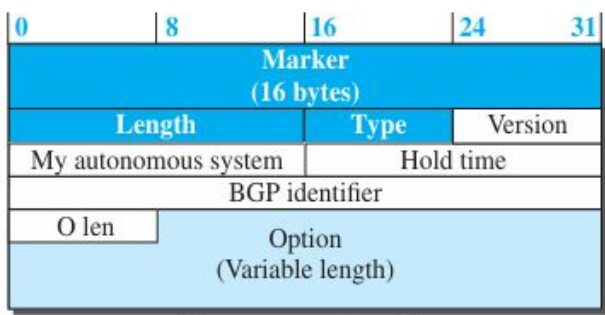
Figure 20.30 Flow diagram for route selection



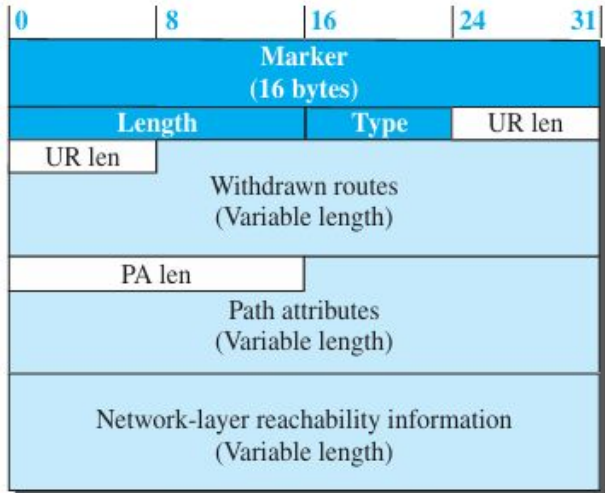
Messages

- Open Message.
- Update Message.
- Keepalive Message.
- Notification.

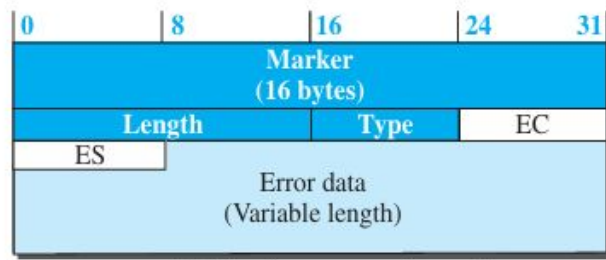
Figure 20.31 BGP messages



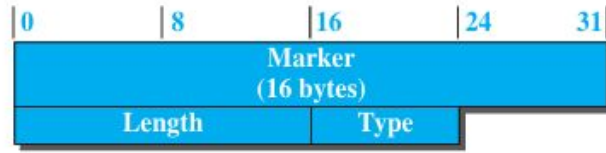
Open message (type 1)



Update message (type 2)



Notification message (type 3)



Keepalive message (type 4)

- Performance

Fields in common header

Marker: Reserved for authentication
Length: Length of total message in bytes
Type: Type of message (1 to 4)

Abbreviations

O len: Option length
EC: Error code
ES: Error subcode
UR len: Unfeasible route length
PA len: Path attribute length

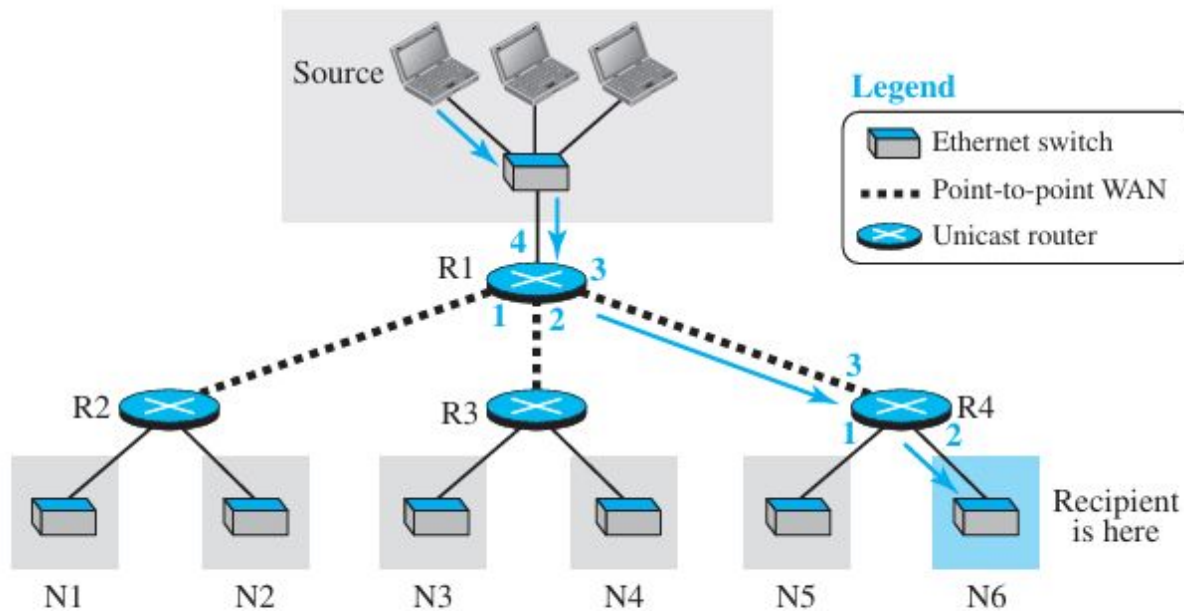
3.3. Multicast

3.3.1. Introduction

- Communication in the Internet today is not only unicasting; multicasting communication is growing fast.
- Here, we first discuss the general ideas behind unicasting, multicasting, and broadcasting. We then talk about some basic issues in multicast routing. Finally, we discuss multicasting routing protocols in the Internet.
- Till now we have learned that the forwarding of a datagram by a router is normally based on the prefix of the destination address in the datagram, which defines the network to which the destination host is connected. Understanding the above forwarding principle, we can now define unicasting, multicasting, and broadcasting.

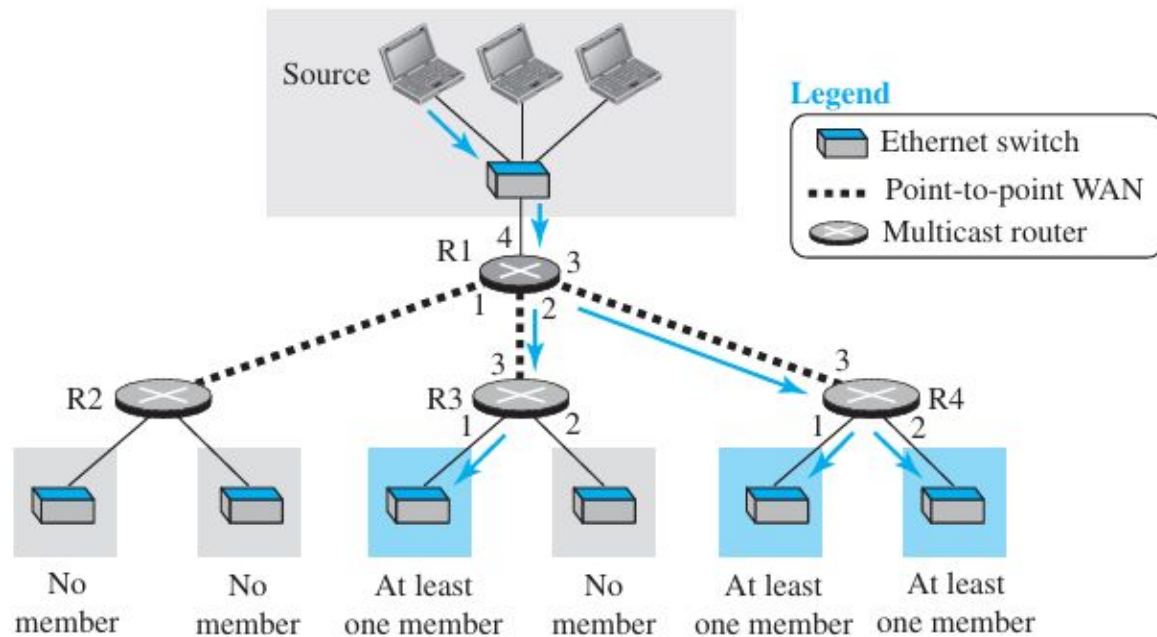
3.3.1.1. Unicast

Figure 21.1 *Unicasting*



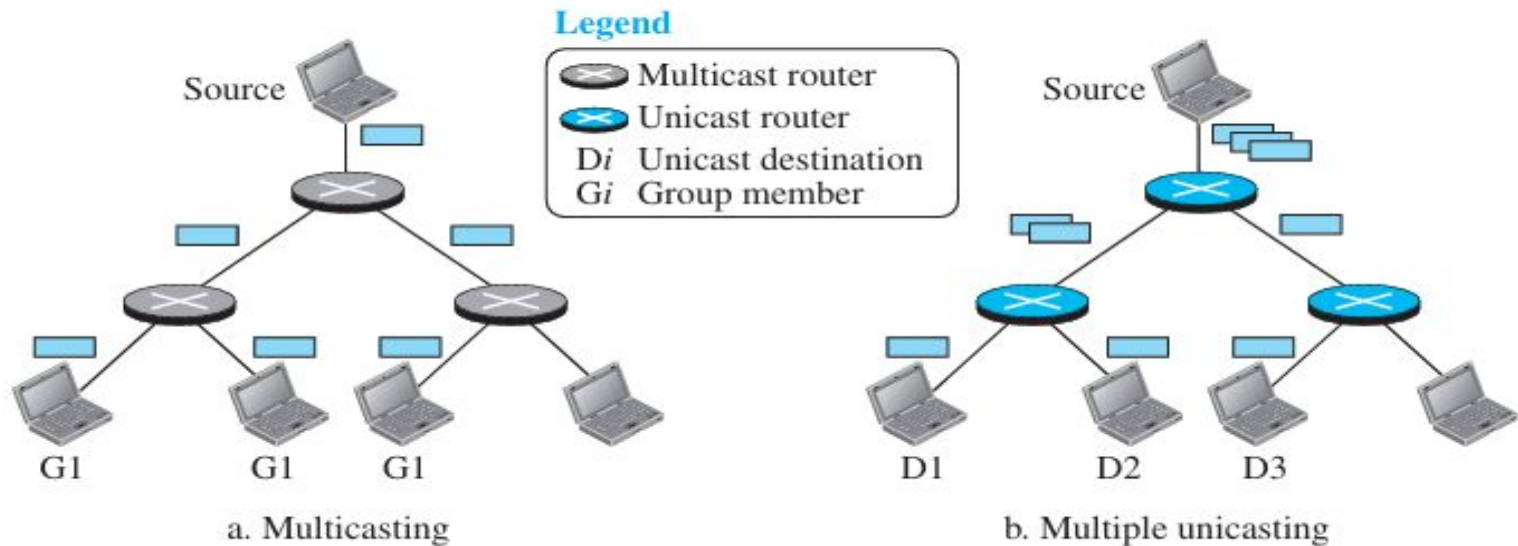
3.3.1.2. Multicasting

Figure 21.2 *Multicasting*



Multicasting versus Multiple Unicasting

Figure 21.3 *Multicasting versus multiple unicasting*



Emulation of Multicasting with Unicasting

- Multicasting is more efficient than multiple unicasting. In Figure 21.3, we can see how multicasting requires less bandwidth than multiple unicasting. In multiple unicasting, some of the links must handle several copies.
- In multiple unicasting, the packets are created by the source with a relative delay between packets. If there are 1,000 destinations, the delay between the first and the last packet may be unacceptable. In multicasting, there is no delay because only one packet is created by the source.

Multicast Applications

- **Access to Distributed Databases.**
- **Information Dissemination.**
- **Teleconferencing.**
- **Distance Learning.**

3.3.1.3. Broadcasting

3.4. MOSPF

3.4. MOSPF

Figure 21.13 Example of tree formation in MOSPF

